

- **Android App Development:** Kotlin is an official language for Android app development and offers various benefits such as ease of learning, concise code, and high performance.
- **Web Development:** Kotlin can compile to JavaScript, making it suitable for frontend web development. It is also designed to work well for server-side development.

Overall, Kotlin's flexibility and compatibility with various platforms make it a popular choice among developers for a wide range of applications.

Kotlin Environment setup for Command Line

To set up a command-line environment for Kotlin, you can follow these steps:

1. **Download and install the Java Development Kit (JDK):** As Kotlin runs on the Java virtual machine, it requires the JDK to be installed on the computer. The latest version can be downloaded from Oracle's official website.
2. **Download the Kotlin compiler:** You can download the latest version of the Kotlin compiler from the official Kotlin website.
3. **Set up the environment variable:** Add the path of the Kotlin compiler to the PATH environment variable. This allows you to run Kotlin commands from anywhere in the command line.
4. **Verify the installation:** To confirm that Kotlin has been installed correctly, open the command line and run a command to display the version of Kotlin that is installed.

Before installing Kotlin, it is necessary to install the Java Development Kit (JDK) and set the path in the local system environment variable since Kotlin runs on the Java Virtual Machine. Refer to this article, "Setting up the environment in Java," to install the JDK and JRE in your system and set the path in the environment variable. After installing the JDK, you can download the latest version of the Kotlin compiler from the official Github Releases page. The latest version as of now is 1.3.31. To set up the Kotlin compiler for the command line, follow these steps:

1. Extract the downloaded Kotlin compiler file to a location where you have write access.
2. Copy the path to the bin directory of kotlin.
3. Open the "Advanced System Settings" in "My Computer Properties" and click on "Environment Variables."
4. Find the "Path" variable in "System Variables" section and click the "Edit" button.